IGEM 2026 / COMPUTATIONAL BIOLOGY + MACHINE LEARNING

Evidence-Grounded Enzyme-Reaction Discovery

From specialist screening to open-universe ranking, verified scientific agents and graph evidence

NJU-China Dry Lab

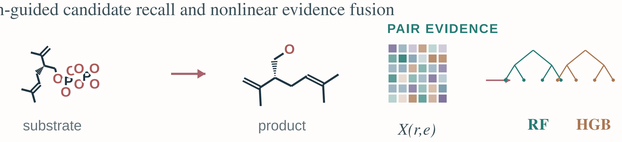
01 | PROGRESSION OF THE RETRIEVAL HYPOTHESIS

SPECIALIST → MULTI-EXPERT → GENERALIZED MODEL

TPS specialist

domain module / retained

Reaction-guided candidate recall and nonlinear evidence fusion

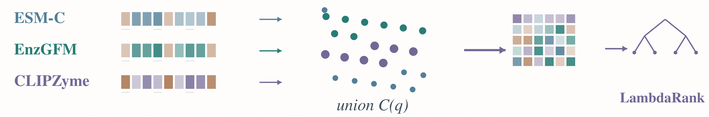


OOF cross-fitting; top-5 / top-10 / top-20 rescue budgets · authentic TPS catalog input

BiME-Rank

retired intermediate model

Independent expert retrieval → union → learned prefix ranking



R2E: LambdaRank · E2R: anchored LambdaMART · expert scores, ranks and missingness masks

Toward an open biochemical universe

Two query directions share one geometry; local evidence has bounded authority.



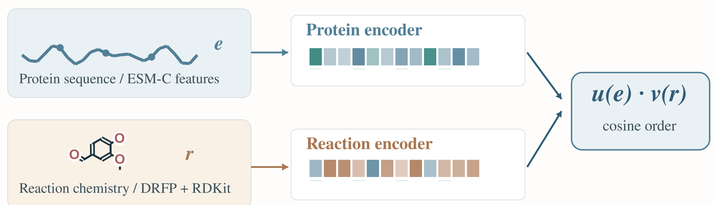
R2E 185,918 proteins **E2R** 11,081 reactions both ranked end-to-end

02 | BRIDGE Broad Retrieval with Inference-Driven Gated Experts

PRIMARY COMPUTATIONAL MODEL

A / SHARED SCORING GEOMETRY

A complete base order for every registered entity

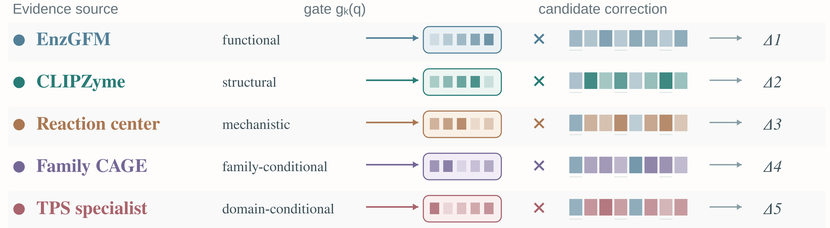


No expert support? Keep the Broad score and complete candidate coverage.

Chemical illustration: real catalog entry RHEA:10036; not a predicted result.

B / INFERENCE-DRIVEN SPECIALISTS

Availability × applicability × query authority



Qualified experts modify an eligible prefix; missing evidence adds exactly zero.

C / TWO MEMORIES, ONE RANKED OUTPUT

Verified relations carry query-specific influence



E2R RELATION READOUT
16 annotated proteins + 8 reaction prototypes

EPISODIC UPDATE
Frozen support attention; confirmed positives

SCOPE
Bounded ranking rights

$$S_{\mathrm{BRIDGE}}(q,c) = S_{\mathrm{Broad}}(q,c) + \alpha_{\mathrm{LT}}(q)\,\Delta_{\mathrm{LT}}(q,c) + \alpha_{\mathrm{epi}}(q,S)\,\Delta_{\mathrm{epi}}(q,c) + \sum_k g_k(q)\,\Delta_k(q,c)$$

COMPLETE BASE ORDER
Global recall retained

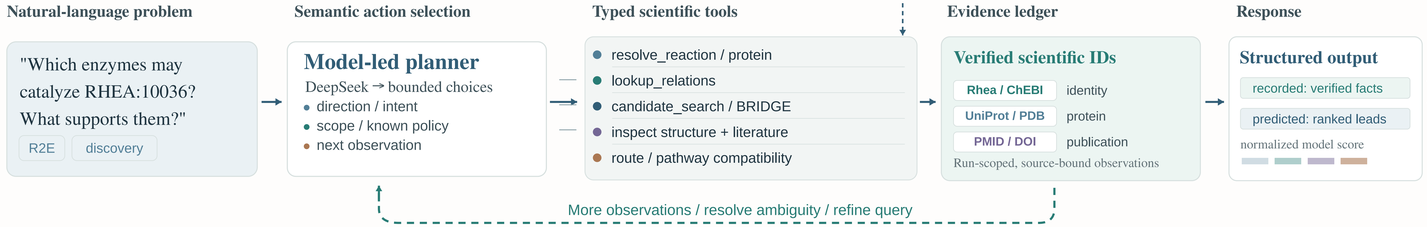
LOCAL RANKING RIGHTS
Prefix / candidate scope

FROZEN + EPISODIC
Query-based authority

03 | COMPASS Conversational Orchestration for Molecular Pathway and Enzyme Search System

BRIDGE scoring

AGENT RUNTIME



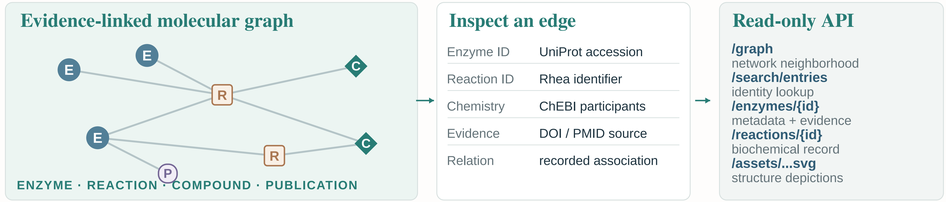
WHAT IS CONSTRAINED
Typed tool contracts · server-verified identifiers · recorded / predicted separation

WHAT REMAINS FLEXIBLE
Multi-step tool choice, optional evidence modules and grounded prose

04 | EDGE Enzyme Dataset and Graph Explorer

VERIFIED DATA LAYER

API



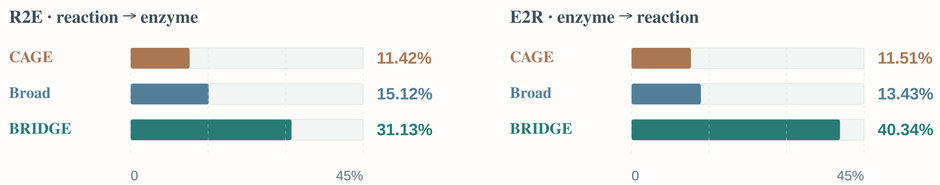
EDGE exposes curated enzyme–reaction associations and provenance-linked records.

COMPASS reads its local backend; predictions retain a separate hypothesis status.

05 | UNIFIED FORMAL EVALUATION

21,505 TRAINING-UNSEEN POSITIVE PAIRS · BOTH RETRIEVAL DIRECTIONS

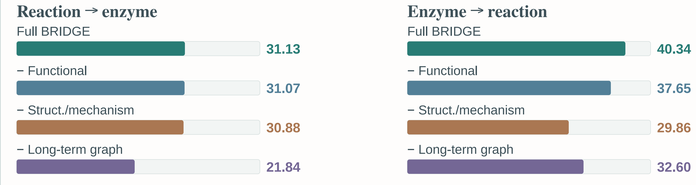
Balanced Hit@10 across complete candidate spaces



Direct Hit@10: R2E CAGE 0.99% · Broad 2.22% · BRIDGE 4.05%

E2R CAGE 9.74% · Broad 7.62% · BRIDGE 25.50%

Expert-group ablation · Balanced Hit@10



Family/domain removal: R2E 31.11%; E2R 40.34% (same as full).

Main-test BRIDGE scores

Task	Averaging	MRR	Hit@10	Hit@100
R2E	Direct	0.026	4.05%	8.04%
R2E	Balanced	0.204	31.13%	40.56%
E2R	Direct	0.150	25.50%	45.42%
E2R	Balanced	0.289	40.34%	60.69%

Train 218,537 / val 5,216 / test 21,505

Filtered full-universe ranking; Balanced weights novelty + degree strata.

BRIDGE: 185,918 proteins / 11,081 reactions · training graph read-only · expert corrections preserve complete base ordering